
EE 708 : Information Theory and Coding

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Homework: Asymptotic Equipartition Property

Please Attempt the following problems from Chapter 3 of “Elements of Information Theory”, by Cover and Thomas, (Second Edition).

3.3, 3.4, 3.6, 3.7, 3.8, 3.9, 3.12

Scanned Copy is attached with the document.

for any $\epsilon > 0$,

$$\Pr \{|Y - \mu| > \epsilon\} \leq \frac{\sigma^2}{\epsilon^2}. \quad (3.32)$$

- (c) (*Weak law of large numbers*) Let $Z_1, Z_2, \dots, Z_n$ be a sequence of i.i.d. random variables with mean μ and variance σ^2 . Let $\bar{Z}_n = \frac{1}{n} \sum_{i=1}^n Z_i$ be the sample mean. Show that

$$\Pr \{|\bar{Z}_n - \mu| > \epsilon\} \leq \frac{\sigma^2}{n\epsilon^2}. \quad (3.33)$$

Thus, $\Pr \{|\bar{Z}_n - \mu| > \epsilon\} \rightarrow 0$ as $n \rightarrow \infty$. This is known as the *weak law of large numbers*.

- 3.2** *AEP and mutual information.* Let (X_i, Y_i) be i.i.d. $\sim p(x, y)$. We form the log likelihood ratio of the hypothesis that X and Y are independent vs. the hypothesis that X and Y are dependent. What is the limit of

$$\frac{1}{n} \log \frac{p(X^n)p(Y^n)}{p(X^n, Y^n)}?$$

- 3.3** *Piece of cake.*

A cake is sliced roughly in half, the largest piece being chosen each time, the other pieces discarded. We will assume that a random cut creates pieces of proportions

$$P = \begin{cases} (\frac{2}{3}, \frac{1}{3}) & \text{with probability } \frac{3}{4} \\ (\frac{2}{5}, \frac{3}{5}) & \text{with probability } \frac{1}{4} \end{cases}$$

Thus, for example, the first cut (and choice of largest piece) may result in a piece of size $\frac{3}{5}$. Cutting and choosing from this piece might reduce it to size $(\frac{3}{5})(\frac{2}{3})$ at time 2, and so on. How large, to first order in the exponent, is the piece of cake after n cuts?

- 3.4** *AEP.* Let X_i be iid $\sim p(x)$, $x \in \{1, 2, \dots, m\}$. Let $\mu = EX$ and $H = -\sum p(x) \log p(x)$. Let $A^n = \{x^n \in \mathcal{X}^n : |-\frac{1}{n} \log p(x^n) - H| \leq \epsilon\}$. Let $B^n = \{x^n \in \mathcal{X}^n : |\frac{1}{n} \sum_{i=1}^n X_i - \mu| \leq \epsilon\}$.

(a) Does $\Pr\{X^n \in A^n\} \rightarrow 1$?

(b) Does $\Pr\{X^n \in A^n \cap B^n\} \rightarrow 1$?

(c) Show that $|A^n \cap B^n| \leq 2^{n(H+\epsilon)}$ for all n .

(d) Show that $|A^n \cap B^n| \geq \left(\frac{1}{2}\right) 2^{n(H-\epsilon)}$ for n sufficiently large.

3.5 *Sets defined by probabilities.* Let $X_1, X_2, \dots$ be an i.i.d. sequence of discrete random variables with entropy $H(X)$. Let

$$C_n(t) = \{x^n \in \mathcal{X}^n : p(x^n) \geq 2^{-nt}\}$$

denote the subset of n -sequences with probabilities $\geq 2^{-nt}$.

(a) Show that $|C_n(t)| \leq 2^{nt}$.

(b) For what values of t does $P(\{X^n \in C_n(t)\}) \rightarrow 1$?

3.6 *AEP-like limit.* Let $X_1, X_2, \dots$ be i.i.d. drawn according to probability mass function $p(x)$. Find

$$\lim_{n \rightarrow \infty} (p(X_1, X_2, \dots, X_n))^{\frac{1}{n}}.$$

3.7 *AEP and source coding.* A discrete memoryless source emits a sequence of statistically independent binary digits with probabilities $p(1) = 0.005$ and $p(0) = 0.995$. The digits are taken 100 at a time and a binary codeword is provided for every sequence of 100 digits containing three or fewer 1's.

(a) Assuming that all codewords are the same length, find the minimum length required to provide codewords for all sequences with three or fewer 1's.

(b) Calculate the probability of observing a source sequence for which no codeword has been assigned.

(c) Use Chebyshev's inequality to bound the probability of observing a source sequence for which no codeword has been assigned. Compare this bound with the actual probability computed in part (b).

3.8 *Products.*

Let

$$X = \begin{cases} 1, & \text{with probability } \frac{1}{2} \\ 2, & \text{with probability } \frac{1}{4} \\ 3, & \text{with probability } \frac{1}{4} \end{cases}$$

Let $X_1, X_2, \dots$ be drawn i.i.d. according to this distribution. Find the limiting behavior of the product

$$(X_1 X_2 \cdots X_n)^{\frac{1}{n}}.$$

3.9 AEP. Let $X_1, X_2, \dots$ be independent, identically distributed random variables drawn according to the probability mass function $p(x), x \in \{1, 2, \dots, m\}$. Thus, $p(x_1, x_2, \dots, x_n) = \prod_{i=1}^n p(x_i)$. We know that $-\frac{1}{n} \log p(X_1, X_2, \dots, X_n) \rightarrow H(X)$ in probability. Let $q(x_1, x_2, \dots, x_n) = \prod_{i=1}^n q(x_i)$, where q is another probability mass function on $\{1, 2, \dots, m\}$.

(a) Evaluate $\lim -\frac{1}{n} \log q(X_1, X_2, \dots, X_n)$, where $X_1, X_2, \dots$ are i.i.d. $\sim p(x)$.

(b) Now evaluate the limit of the log likelihood ratio $\frac{1}{n} \log \frac{q(X_1, \dots, X_n)}{p(X_1, \dots, X_n)}$ when $X_1, X_2, \dots$ are i.i.d. $\sim p(x)$. Thus, the odds favoring q are exponentially small when p is true.

3.10 Random box size.

An n -dimensional rectangular box with sides $X_1, X_2, X_3, \dots, X_n$ is to be constructed. The volume is $V_n = \prod_{i=1}^n X_i$. The edge length l of a n -cube with the same volume as the random box is $l = V_n^{1/n}$. Let $X_1, X_2, \dots$ be i.i.d. uniform random variables over the unit interval $[0, 1]$. Find $\lim_{n \rightarrow \infty} V_n^{1/n}$ and compare to $(E V_n)^{1/n}$. Clearly, the expected edge length does not capture the idea of the volume of the box. The geometric mean, rather than the arithmetic mean, characterizes the behavior of products.

3.11 Proof of Theorem 3.3.1. This problem shows that the size of the smallest “probable” set is about 2^{nH} . Let $X_1, X_2, \dots, X_n$ be i.i.d. $\sim p(x)$. Let $B_\delta^{(n)} \subset \mathcal{X}^n$ such that $\Pr(B_\delta^{(n)}) > 1 - \delta$. Fix $\epsilon < \frac{1}{2}$.

(a) Given any two sets A, B such that $\Pr(A) > 1 - \epsilon_1$ and $\Pr(B) > 1 - \epsilon_2$, show that $\Pr(A \cap B) > 1 - \epsilon_1 - \epsilon_2$. Hence, $\Pr(A_\epsilon^{(n)} \cap B_\delta^{(n)}) \geq 1 - \epsilon - \delta$.

(b) Justify the steps in the chain of inequalities

$$1 - \epsilon - \delta \leq \Pr(A_\epsilon^{(n)} \cap B_\delta^{(n)}) \quad (3.34)$$

$$= \sum_{A_\epsilon^{(n)} \cap B_\delta^{(n)}} p(x^n) \quad (3.35)$$

$$\leq \sum_{A_\epsilon^{(n)} \cap B_\delta^{(n)}} 2^{-n(H-\epsilon)} \quad (3.36)$$

$$= |A_\epsilon^{(n)} \cap B_\delta^{(n)}| 2^{-n(H-\epsilon)} \quad (3.37)$$

$$\leq |B_\delta^{(n)}| 2^{-n(H-\epsilon)}. \quad (3.38)$$

(c) Complete the proof of the theorem.

3.12 *Monotonic convergence of the empirical distribution.*

Let $\hat{p}_n$ denote the empirical probability mass function corresponding to $X_1, X_2, \dots, X_n$ i.i.d. $\sim p(x)$, $x \in \mathcal{X}$. Specifically,

$$\hat{p}_n(x) = \frac{1}{n} \sum_{i=1}^n I(X_i = x)$$

is the proportion of times that $X_i = x$ in the first n samples, where I is the indicator function.

(a) Show for $\mathcal{X}$ binary that

$$ED(\hat{p}_{2n} \parallel p) \leq ED(\hat{p}_n \parallel p).$$

Thus, the expected relative entropy “distance” from the empirical distribution to the true distribution decreases with sample size. (Hint: Write $\hat{p}_{2n} = \frac{1}{2}\hat{p}_n + \frac{1}{2}\hat{p}'_n$ and use the convexity of D .)

(b) Show for an arbitrary discrete $\mathcal{X}$ that

$$ED(\hat{p}_n \parallel p) \leq ED(\hat{p}_{n-1} \parallel p).$$

(Hint: Write $\hat{p}_n$ as the average of n empirical mass functions with each of the n samples deleted in turn.)

3.13 *Calculation of typical set.* To clarify the notion of a typical set $A_\epsilon^{(n)}$ and the smallest set of high probability $B_\delta^{(n)}$, we will calculate the set for a simple example. Consider a sequence of i.i.d. binary random variables, $X_1, X_2, \dots, X_n$, where the probability that $X_i = 1$ is 0.6 (and therefore the probability that $X_i = 0$ is 0.4).

(a) Calculate $H(X)$.

(b) With $n = 25$ and $\epsilon = 0.1$, which sequences fall in the typical set $A_\epsilon^{(n)}$? What is the probability of the typical set? How many elements are there in the typical set? (This involves computation of a table of probabilities for sequences with k 1's, $0 \leq k \leq 25$, and finding those sequences that are in the typical set.)

(c) How many elements are there in the smallest set that has probability 0.9?

(d) How many elements are there in the intersection of the sets in parts (b) and (c)? What is the probability of this intersection?